

Tensile test data for two specimens of AISI 1018 steel

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Tensile test.

Tensile testing was performed in accordance with ASTM E 8–04 Standard Test Methods for Tension Testing of Metallic Materials on a U.S.-made MTS-810 universal machine. The load capacity of the machine is 250 kN. The hydraulically operated wedge-shaped grip ensures that the specimens are clamped. The control software for the machine is the MTS FlexTest 40.

The specimen was immobilized on the fixed head of the machine, while on the moving head an offset was applied at a speed of 0.1 mm/s. ASTM E 8-04 states that repetitibility and reproducibility studies of a test are highly recommended. To do this, two or three measurements must be made on test specimens. In this case, two specimens were studied, which have the geometric configuration shown in Figure 1.

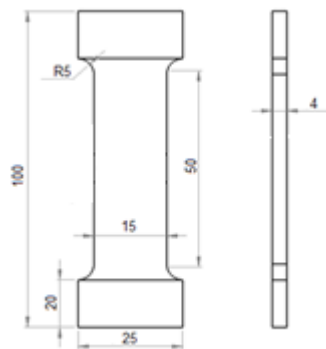


Figure 1. Geometric shape and dimensions of the tensile specimens.

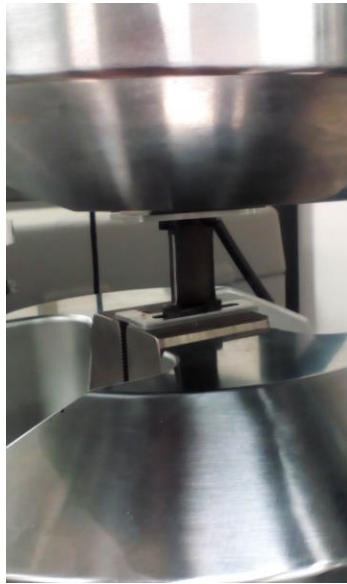


Figure 2. Tensile specimen.

Figure 3 shows the stress-strain graphs obtained from the tests carried out on the two specimens of the base material, which are denoted as MB1 and MB2.

In Table 1, in the row "Plasticity" refers to the parameters "Resistance Coefficient" K and "Deformation Hardening Exponent" n to adjust the relationship between stress σ and strain ε in the plastic zone through a Hollomon-type equation (Equation 1).

$$\sigma = K\varepsilon^n \quad (1)$$

Table 1. Mechanical properties of the AISI 1015 steel used in this work.

Specimen	Yield limit σ_y (MPa)	Ultimate stress σ_u (MPa)	Plasticity	
			K	N
MB1	337,98	429,86	652	0,14
MB2	335,11	432,50	627,11	0,118
Average	336,55	431,18	639,555	0,129

Figure 3 shows the stress-strain curves experimentally determined for the steel studied. In the attached file "data S-e MB.xlsx" you will find the data that were used to construct the two curves shown in Figure 3.

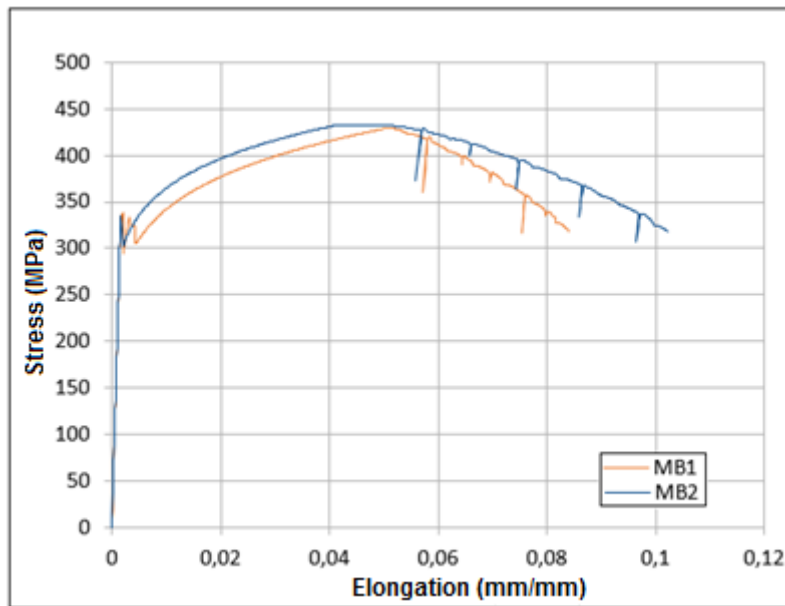


Figure 3. Stress-strain curves of the AISI 1015 steel used in this work.

Conclusions

The tensile test was carried out on two AISI 1015 steel specimens. This made it possible to determine mechanical properties such as yield limit σ_y , ultimate stress σ_u and plasticity parameters K and n .

References

Standard ASTM E 8–04 Standard Test Methods for Tension Testing of Metallic Materials (2004).